

EXPERIMENT 10

Implement Boundary condition between two different dielectric medium in Magnetic Field And find the relative permeability of the different medium.

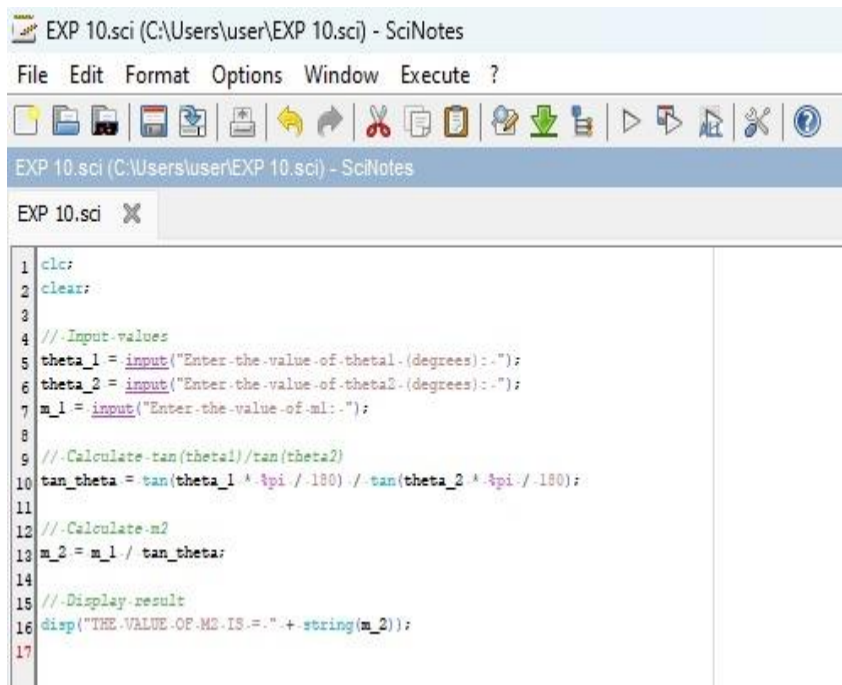
Aim:

To write program for Boundary condition between two different dielectric media in magnetic field using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:



```
EXP 10.sci (C:\Users\user\EXP 10.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 10.sci (C:\Users\user\EXP 10.sci) - SciNotes
EXP 10.sci X
1 clc;
2 clear;
3
4 //Input-values
5 theta_1 = input("Enter the value of theta1 (degrees):-");
6 theta_2 = input("Enter the value of theta2 (degrees):-");
7 m_1 = input("Enter the value of m1:-");
8
9 //Calculate tan(theta1)/tan(theta2)
10 tan_theta = tan(theta_1 * %pi / 180) / tan(theta_2 * %pi / 180);
11
12 //Calculate m2
13 m_2 = m_1 / tan_theta;
14
15 //Display result
16 disp('THE VALUE OF M2 IS:.' + string(m_2));
17
```

Output:

Enter the value of theta1 (degrees): 50

Enter the value of theta2 (degrees): 75

Enter the value of μ_1 : 1×10^{-3}

"THE VALUE OF μ_2 IS = 21.920937"

Theoretical calculation:

Theoretical Calculation:

Magnetic boundary condition

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

$$\mu_2 = \mu_1 \frac{\tan \theta_2}{\tan \theta_1}$$

$$\mu_2 = (1 \times 10^{-3}) \frac{\tan 45^\circ}{\tan 50^\circ}$$

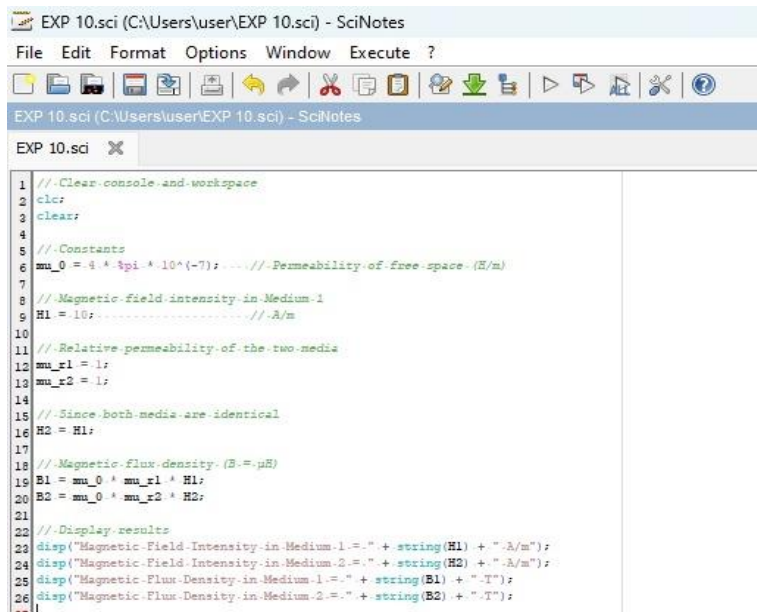
$$\mu_2 = (1 \times 10^{-3}) \frac{1}{1.918}$$

$$\mu_2 = 8.39 \times 10^{-4}$$

$$\text{Theoretical value } \mu_2 = 0.000839$$

Case 1: Consider two identical media in the presence of a magnetic field. The relative permeability in Both media is 1. What will be the effect of this on the magnetic field and flux density across the Boundary?

Program:

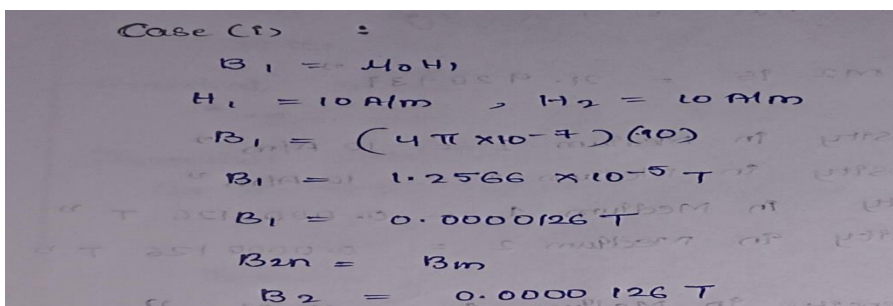


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EXP 10.sci (C:\Users\user\EXP 10.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 10.sci (C:\Users\user\EXP 10.sci) - SciNotes
EXP 10.sci ✕
1 // Clear console and workspace
2 clc;
3 clear;
4
5 // Constants
6 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
7
8 // Magnetic field intensity in Medium 1
9 H1 = 10; // A/m
10
11 // Relative permeability of the two media
12 mu_r1 = 1;
13 mu_r2 = 1;
14
15 // Since both media are identical
16 H2 = H1;
17
18 // Magnetic flux density (B = μH)
19 B1 = mu_0 * mu_r1 * H1;
20 B2 = mu_0 * mu_r2 * H2;
21
22 // Display results
23 disp("Magnetic Field Intensity in Medium 1 = " + string(H1) + " A/m");
24 disp("Magnetic Field Intensity in Medium 2 = " + string(H2) + " A/m");
25 disp("Magnetic Flux Density in Medium 1 = " + string(B1) + " T");
26 disp("Magnetic Flux Density in Medium 2 = " + string(B2) + " T");
```

Output:

```
"Magnetic Field Intensity in Medium 1 = 10 A/m"
"Magnetic Field Intensity in Medium 2 = 10 A/m"
"Magnetic Flux Density in Medium 1 = 0.0000126 T"
"Magnetic Flux Density in Medium 2 = 0.0000126 T"
```

Theoretical calculation:

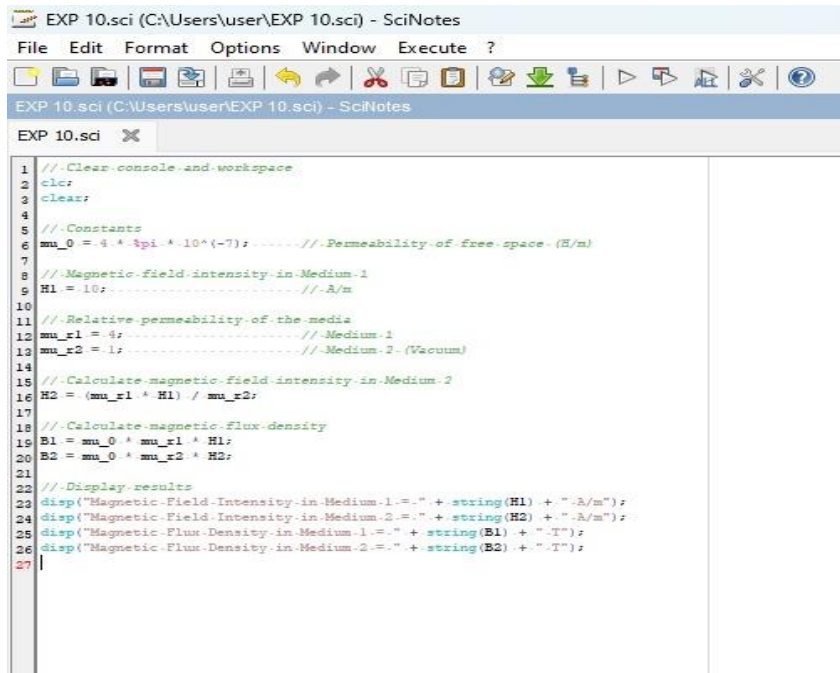


Case (1) :

$$B_1 = \mu_0 H_1$$
$$H_1 = 10 \text{ A/m} \quad , \quad H_2 = 10 \text{ A/m}$$
$$B_1 = (4\pi \times 10^{-7}) (10)$$
$$B_1 = 1.2566 \times 10^{-5} \text{ T}$$
$$B_1 = 0.0000126 \text{ T}$$
$$B_2 = B_1$$
$$B_2 = 0.0000126 \text{ T}$$

Case 2: In a system with two media, where the relative permeability of medium 1 is $\mu_{r1}=4$ and medium 2 is a vacuum ($\mu_{r2}=1$), how would the magnetic field intensity and flux density behave at the boundary?

Program:



```

1 // Clear console and workspace
2 clc;
3 clear;
4
5 // Constants
6 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
7
8 // Magnetic field intensity in Medium 1
9 H1 = 10; // A/m
10
11 // Relative permeability of the media
12 mu_r1 = 4; // Medium 1
13 mu_r2 = 1; // Medium 2 (Vacuum)
14
15 // Calculate magnetic field intensity in Medium 2
16 H2 = (mu_r1 * H1) / mu_r2;
17
18 // Calculate magnetic flux density
19 B1 = mu_0 * mu_r1 * H1;
20 B2 = mu_0 * mu_r2 * H2;
21
22 // Display results
23 disp('Magnetic Field Intensity in Medium 1 = ' + string(H1) + ' A/m');
24 disp('Magnetic Field Intensity in Medium 2 = ' + string(H2) + ' A/m');
25 disp('Magnetic Flux Density in Medium 1 = ' + string(B1) + ' T');
26 disp('Magnetic Flux Density in Medium 2 = ' + string(B2) + ' T');
27

```

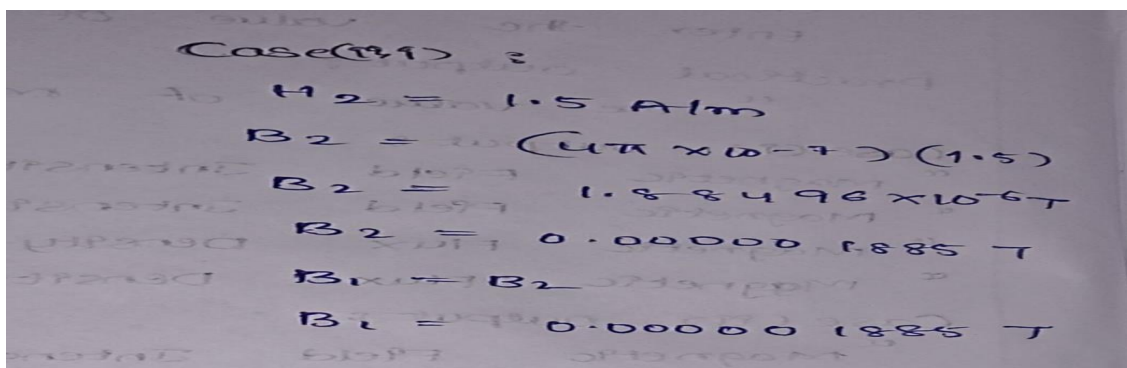
Output:

```

"Magnetic Field Intensity in Medium 1 = 10 A/m"
"Magnetic Field Intensity in Medium 2 = 40 A/m"
"Magnetic Flux Density in Medium 1 = 0.0000503 T"
"Magnetic Flux Density in Medium 2 = 0.0000503 T"

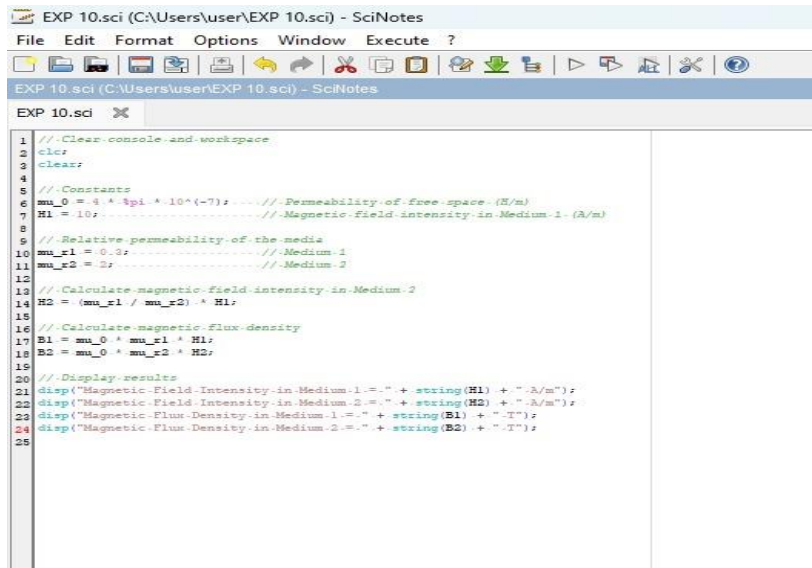
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Theoretical calculation:



Case 3: If medium 1 has a relative permeability $\mu_{r1}=0.3$ and medium 2 has a relative permeability $\mu_{r2}=2$, calculate the magnetic field intensity in both media at the boundary if the magnetic field intensity in medium 1 is known to be $H_1=10$ A/m

Program:



```

1 // Clear console and workspace
2 clear;
3 clc;
4
5 // Constants
6 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
7 H1 = 10; // Magnetic field intensity in Medium 1 (A/m)
8
9 // Relative permeability of the media
10 mu_r1 = 0.3; // Medium 1
11 mu_r2 = 2; // Medium 2
12
13 // Calculate magnetic field intensity in Medium 2
14 H2 = (mu_r1 / mu_r2) * H1;
15
16 // Calculate magnetic flux density
17 B1 = mu_0 * mu_r1 * H1;
18 B2 = mu_0 * mu_r2 * H2;
19
20 // Display results
21 disp("Magnetic Field Intensity in Medium 1 = " + string(H1) + " A/m");
22 disp("Magnetic Field Intensity in Medium 2 = " + string(H2) + " A/m");
23 disp("Magnetic Flux Density in Medium 1 = " + string(B1) + " T");
24 disp("Magnetic Flux Density in Medium 2 = " + string(B2) + " T");
25

```

Output:

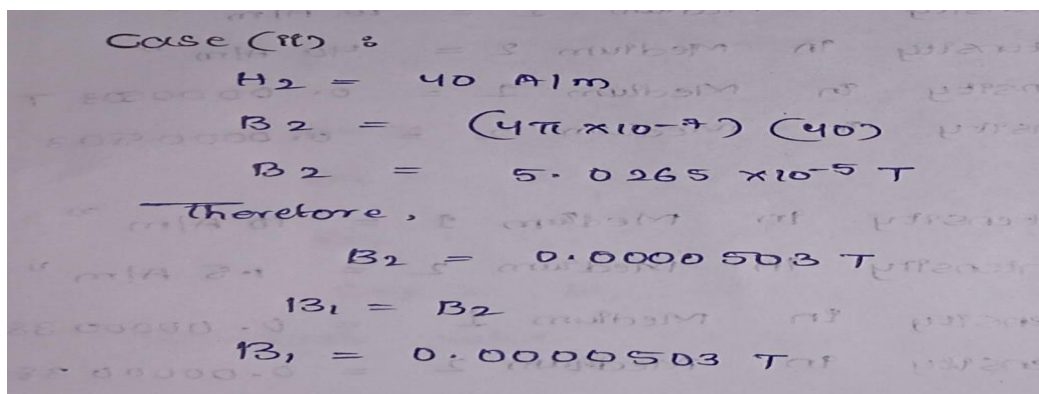
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"Magnetic Field Intensity in Medium 1 = 10 A/m"
"Magnetic Field Intensity in Medium 2 = 1.5 A/m"
"Magnetic Flux Density in Medium 1 = 0.0000038 T"
"Magnetic Flux Density in Medium 2 = 0.0000038 T"

```

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Theoretical calculation:



Case (iii) :

$$H_2 = 40 \text{ A/m}$$

$$B_2 = (4\pi \times 10^{-7}) (40)$$

$$B_2 = 5.0265 \times 10^{-5} \text{ T}$$

Therefore,

$$B_2 = 0.0000503 \text{ T}$$

$$B_1 = B_2$$

$$B_1 = 0.0000503 \text{ T}$$

Result :

Thus the effect of two different media on the magnetic field and flux density across the boundary is calculated.